LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Daily Practice: Ch 1 Chemical Reactions and Equations

Chapter / Topic: Chemical Reactions and Equations · Week 1 of the 90-day plan

For teacher use only. Do not distribute to students.

DAY 1: WRITING AND BALANCING EQUATIONS

Q1. Magnesium ribbon burns with a dazzling white flame and turns into a white powder. The white powder is:

Answer: (d) magnesium oxide: $2Mg + O_2 \rightarrow 2MgO$

Q2. We balance a chemical equation so that it obeys the law of:

Answer: (c) conservation of mass: atoms are neither created nor destroyed in a reaction

Q3. Which state symbol shows that a substance is dissolved in water?

Answer: (c) (aq), aqueous

Q4. Heating limestone ($CaCO_3$) to make quicklime is an example of:

Answer: (b) a thermal decomposition reaction: $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

Q5. What does the arrow (→) in a chemical equation mean? On which side do we write the reactants?

Answer: The arrow means 'gives' or 'forms', and shows the direction of the reaction. Reactants are written on the left, products on the right.

Q6. Balance these equations: (i) $Fe + H_2O \rightarrow Fe_3O_4 + H_2$ (ii) $Na + H_2O \rightarrow NaOH + H_2$ (iii) $Al + O_2 \rightarrow Al_2O_3$ (iv) $KClO_3 \rightarrow KCl + O_2$

Answer: (i) $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$
(ii) $2Na + 2H_2O \rightarrow 2NaOH + H_2$
(iii) $4Al + 3O_2 \rightarrow 2Al_2O_3$
(iv) $2KClO_3 \rightarrow 2KCl + 3O_2$

Q7. Write a balanced equation with state symbols: solid zinc reacts with aqueous copper sulphate to give aqueous zinc sulphate and solid copper.

Answer: $Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$

Q8. Hydrogen gas combines with chlorine gas to form hydrogen chloride gas. Write the balanced equation and name the type of reaction.

Answer: $H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$. It is a combination reaction (two substances form one product).

DAY 2: TYPES OF REACTIONS

Q9. Which of the following is a double displacement reaction?

Answer: (d) $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$: the two compounds exchange ions

Q10. An iron nail is kept in copper sulphate solution for about 20 minutes. The nail:

Answer: (a) gets a brownish coating of copper, because iron displaces copper

Q11. In the reaction $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$, the substance that is reduced is:

Answer: (a) MnO_2: it loses oxygen. (HCl is oxidised: it loses hydrogen.)

Q12. Why is respiration called an exothermic reaction? Write the balanced equation for the breakdown of glucose in our cells.

Answer: Respiration releases energy, so it is exothermic.
$C_6H_{12}O_6(aq) + 6O_2(aq) \rightarrow 6CO_2(aq) + 6H_2O(l) + energy$

Q13. Silver bromide is used in black-and-white photography. Write the balanced equation for what happens when it is exposed to light, and name the type of…

Answer: $2AgBr(s) \rightarrow 2Ag(s) + Br_2(g)$ in light. Photolytic decomposition (decomposition by light). The silver formed makes the image dark.

Q14. Name the type of each reaction and give a reason: (i) $NaOH + HCl \rightarrow NaCl + H_2O$ (ii) $2FeSO_4 \rightarrow Fe_2O_3 + SO_2 + SO_3$ (on heating) (iii) $Mg + 2HCl \rightarrow MgCl_2 + H$…

Answer: (i) Double displacement (neutralisation): the two compounds exchange ions.
(ii) Thermal decomposition: one substance breaks into simpler ones on heating.
(iii) Displacement: the more reactive metal, magnesium, displaces hydrogen from the acid.

Q15. Zinc reacts with dilute hydrochloric acid, but copper does not. Write the balanced equation for the reaction that happens, and explain what this…

Answer: $Zn + 2HCl \rightarrow ZnCl_2 + H_2$. Zinc is more reactive than hydrogen, so it displaces hydrogen from the acid. Copper is less reactive than hydrogen, so it cannot displace it and there is no reaction.

DAY 3: MIXED PRACTICE AND CHALLENGE

Q16. **Assertion (A):** Most decomposition reactions are endothermic. **Reason (R):** Energy, as heat, light or electricity, is needed to break a compound into…

Answer: (a) Both are true, and R explains A.

Q17. Which of these changes involve oxidation? (i) rusting of iron (ii) rancidity of butter (iii) respiration (iv) melting of wax

Answer: (b) (i), (ii) and (iii). Melting of wax is a physical change.

Q18. A student dipped a clean copper wire into colourless silver nitrate solution. After an hour, the solution had turned blue, and a shiny greyish-white…

(i) The deposit is silver. The solution turned blue because copper nitrate formed, and copper salts in solution are blue.
(ii) $Cu(s) + 2AgNO_3(aq) \rightarrow Cu(NO_3)_2(aq) + 2Ag(s)$
(iii) A displacement reaction. Copper is more reactive than silver, so it displaces silver from its salt solution.

Q19. (a) Burning magnesium keeps burning in carbon dioxide: $2Mg + CO_2 \rightarrow 2MgO + C$. Which substance is oxidised and which is reduced? (b) Why is hydrogen…

Answer: (a) Mg is oxidised (gains oxygen to form MgO). CO_2 is reduced (loses oxygen to form carbon).
(b) Light decomposes it: $2H_2O_2 \rightarrow 2H_2O + O_2$. Dark bottles keep light out.
(c) $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ + heat. Exothermic (combustion gives out heat).

Q20. A student adds barium chloride solution to sodium sulphate solution in a test tube. (i) What does she see? (ii) Write the balanced equation with…

Answer: (i) A white precipitate forms.
(ii) $Na_2SO_4(aq) + BaCl_2(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$
(iii) Double displacement (precipitation) reaction. Another example: $AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$, a white precipitate of silver chloride.